

Apostol Chapter 1 - Exercise 2

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If $(a, b) = (a, c) = 1$, then $(a, bc) = 1$

Obviously, $1|a$ and $1|bc$.

Let $e \in \mathbb{Z}$ such that $e|a$ and $e|bc$.

Since $(a, b) = 1$, we can write

$$1 = ax + by \text{ for some integers } x, y,$$

multiplying both sides by c we get

$$c = acx + bcy.$$

From $e|a$ and $e|bc$ we have that there exists $q_1, q_2 \in \mathbb{Z}$ such that

$$a = eq_1 \text{ and } bc = eq_2.$$

Substituting in the previous equation we get

$$c = eq_1cx + eq_2y = e(q_1cx + q_2y),$$

which means that $e|c$. Since $e|a$ and $e|c$ we get that $e|(a, c)$, but $(a, c) = 1$, so $e|1$. Therefore $e = \pm 1$. In particular, $e|1$. Therefore $(a, bc) = 1$